Retrieval-Augmented Generation (RAG) Architecture Report

Section 1: Executive Overview

Retrieval-Augmented Generation (RAG) combines dense semantic retrieval with deep generative AI models.
By indexing external PDF documents into Pinecone vector databases, language models access authoritative facts.
This architecture eliminates common AI hallucinations and grounds responses in verified context.

Section 2: The End-to-End Pipeline Workflow

1. Ingestion & Chunking: Text is extracted and split with LangChain RecursiveCharacterTextSplitter.
2. Tokenization & Embeddings: Chunks are converted into 768-dimensional vectors using Google Gemini.
3. Vector Indexing: Embeddings and metadata are indexed in Pinecone namespaces with user ID isolation.
4. Query Retrieval: When a user asks a question, Pinecone executes cosine similarity to find top matches.
5. Decoding & Synthesis: Gemini Flash analyzes retrieved chunks and synthesizes a grounded answer.

Section 3: Security & Session Isolation

Authentication uses industry-standard JSON Web Tokens (JWT) with bcrypt salt rounds for password safety.
Every conversation is persisted in MongoDB and linked with the authenticated user ID.
All API request inputs are strictly validated using Zod schemas on both frontend and backend.